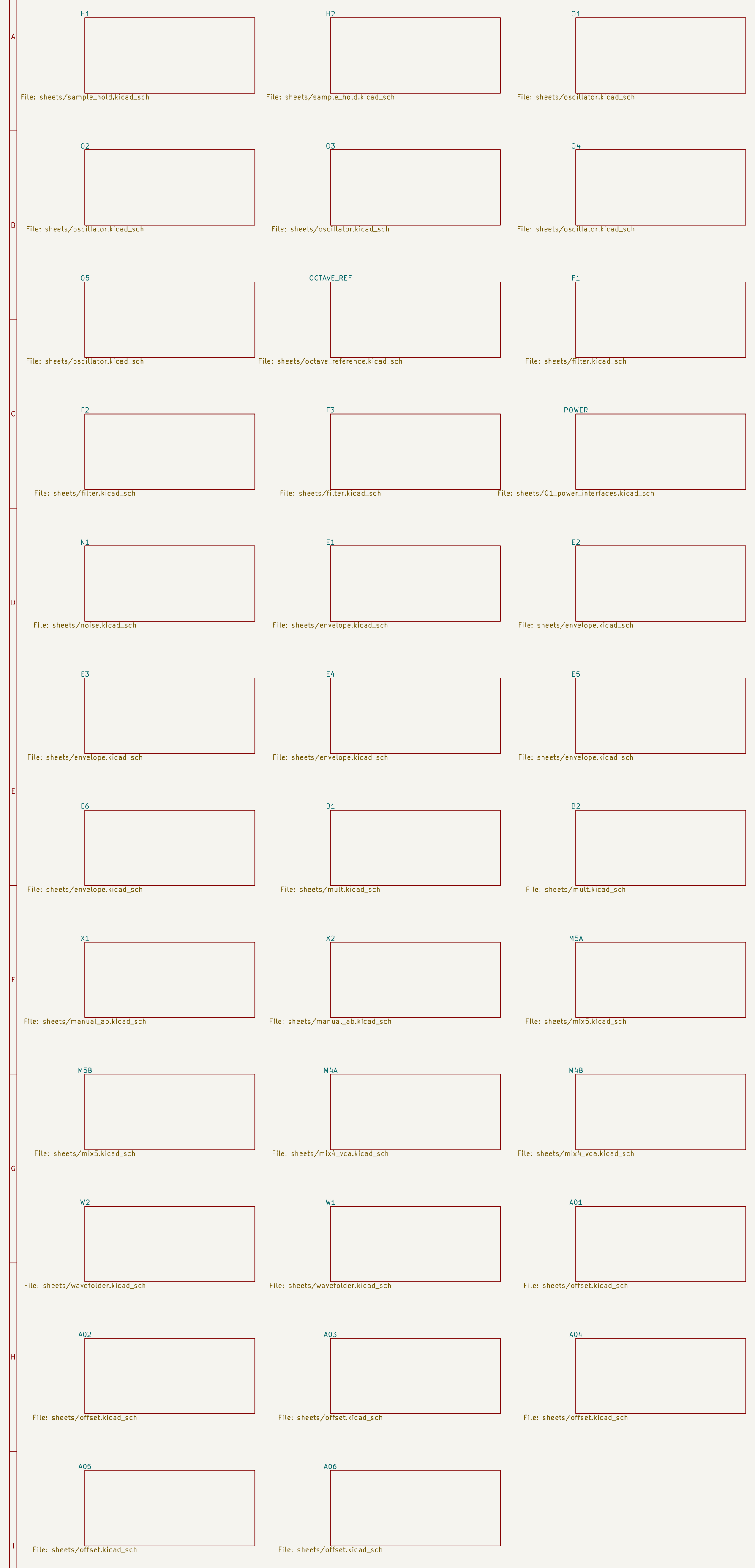
H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

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File: zudo-osc-hole-field.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/36

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-12V
+5V
OPA4197IPWR
OPA4196IDR
SN74HC14DR
ADG5412FBRUZ
LM393BIDR
BAS16GW-QX
APHHS1005QWF/D
CD74HC221M96
BAT54S,215
TRIGGER_TIP
CONTACT
REF_5V
REF_N5V
SLEW_STORAGE
SLEW_BUFFERED
TRIGGER_PROTECTED
TRIGGER_BUFFERED
IN_BUFFERED
IN_PROTECTED
BUTTON_ACTIVE
TRIGGER_GATE
TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
RAW_HELD
OUT_TIP
SLEW_POT_IN

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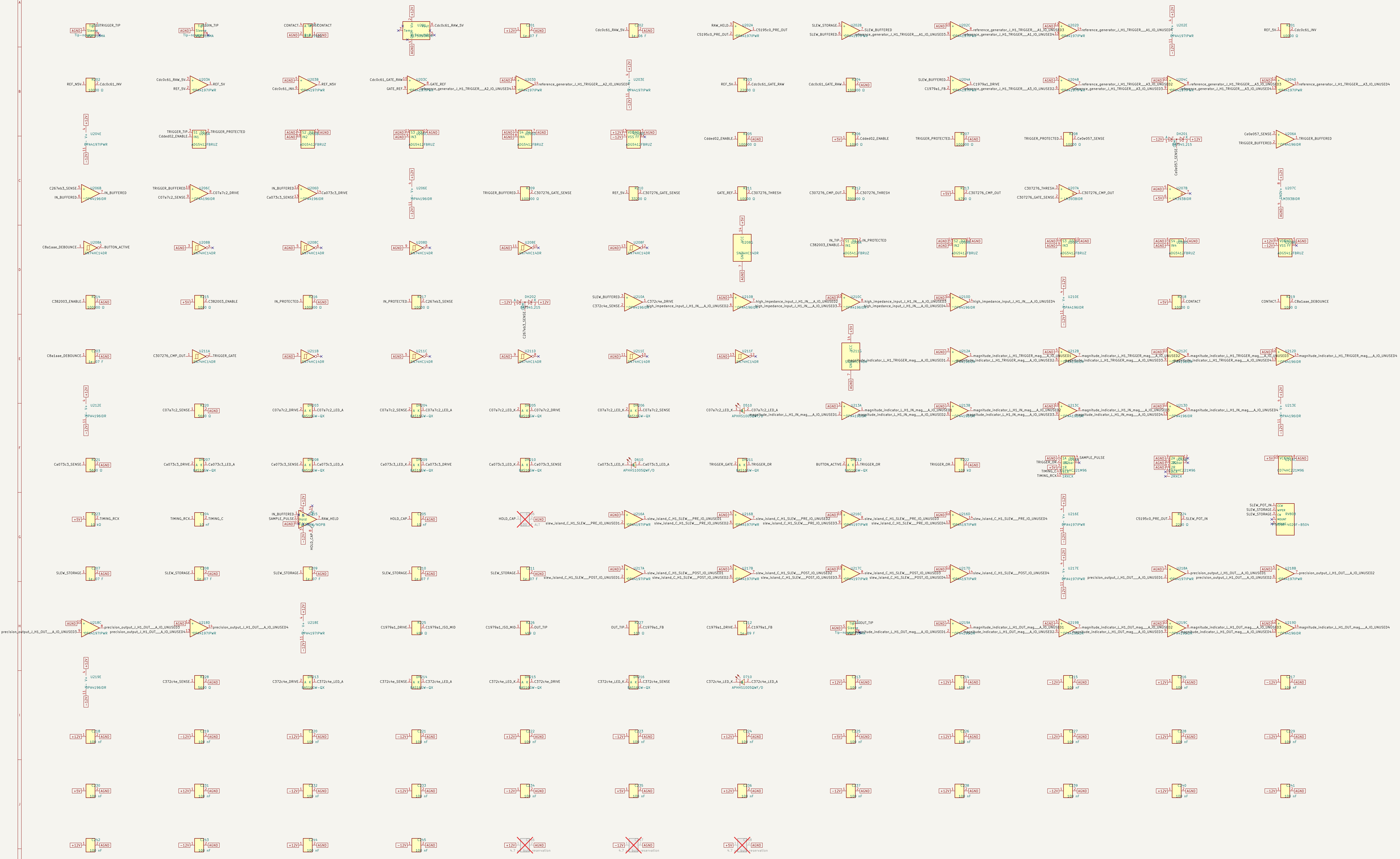
Size: A1 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 2/36



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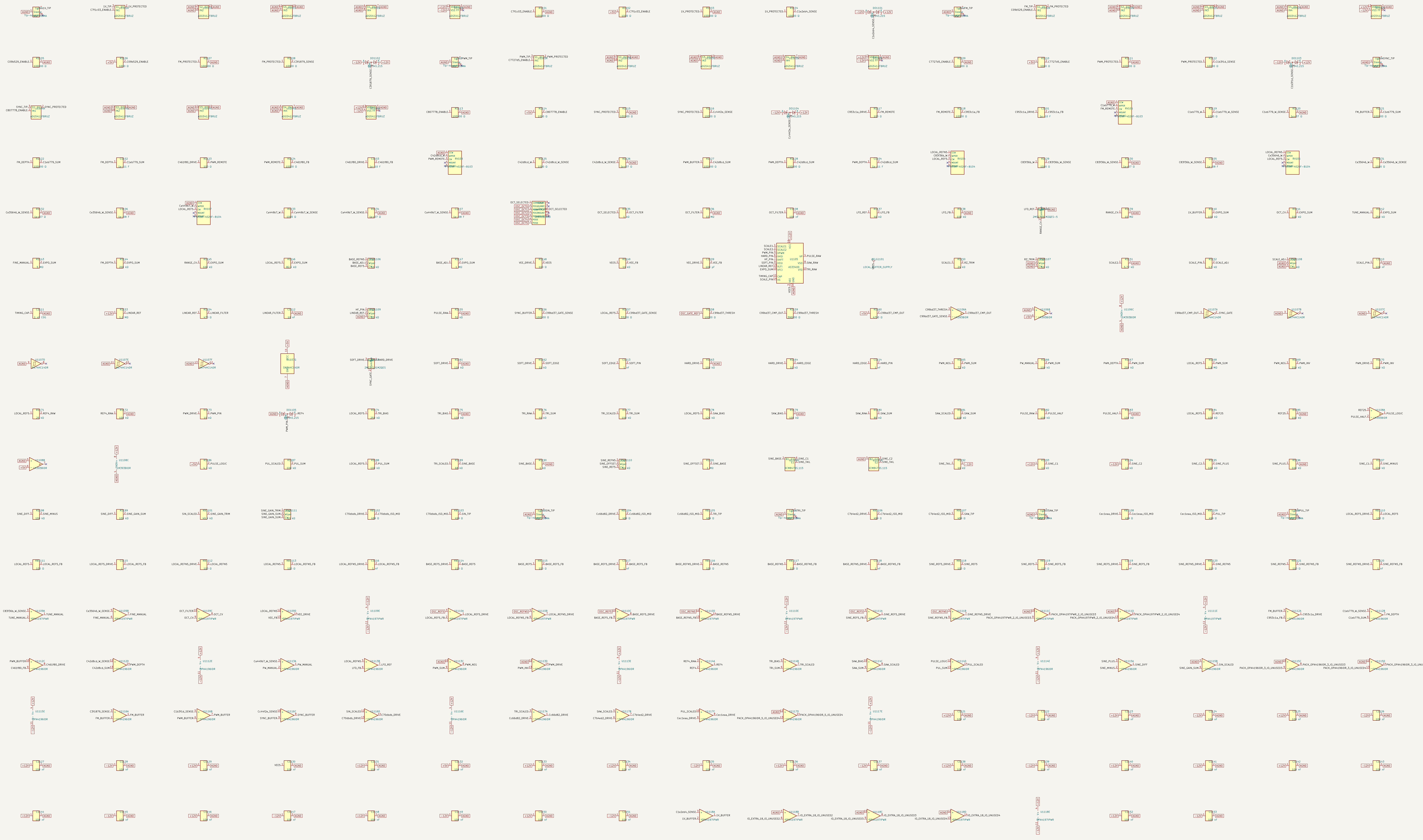
KiCad E.D.A. 10.0.6 | Id: 3/36



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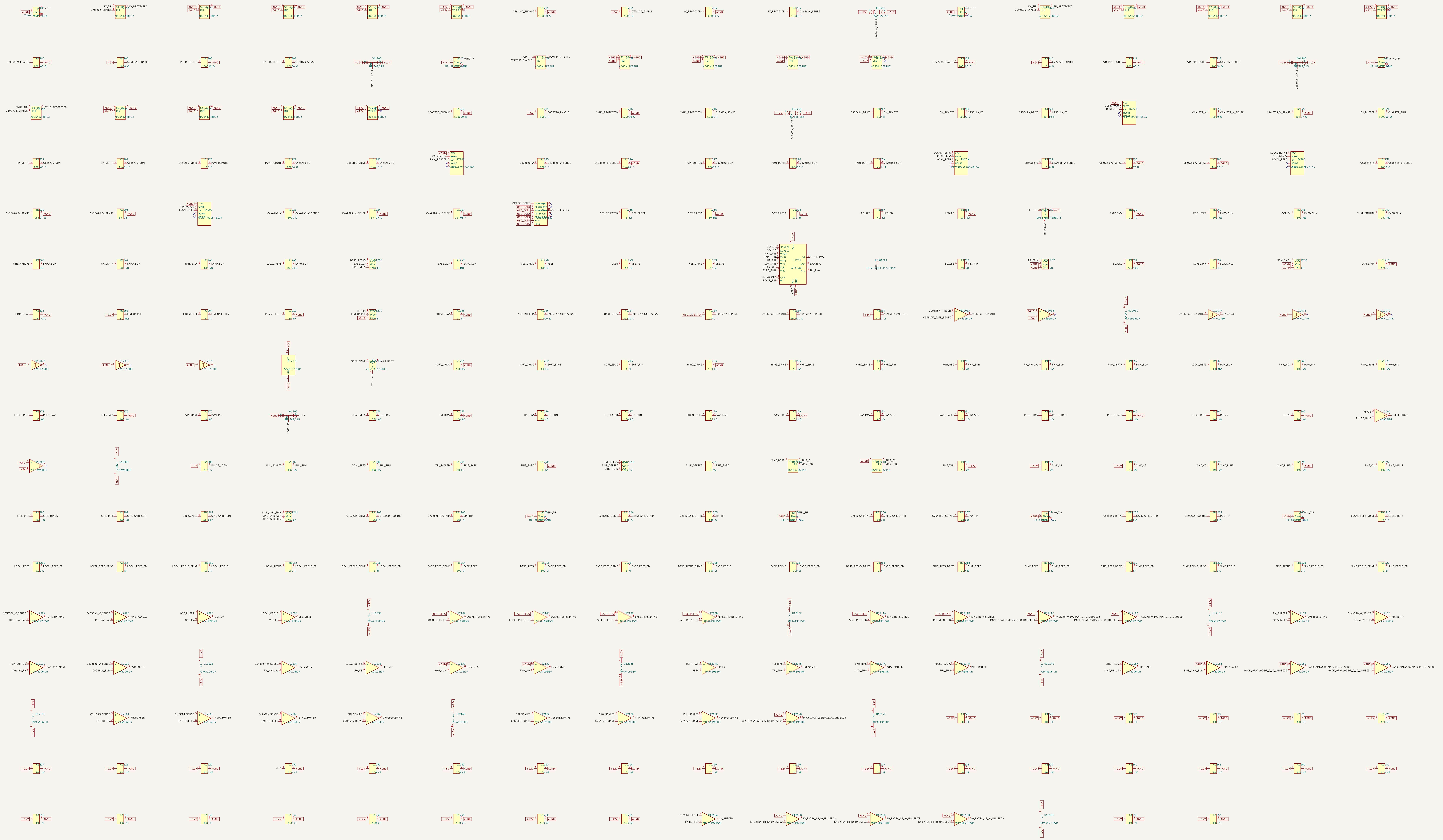
Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 4/36



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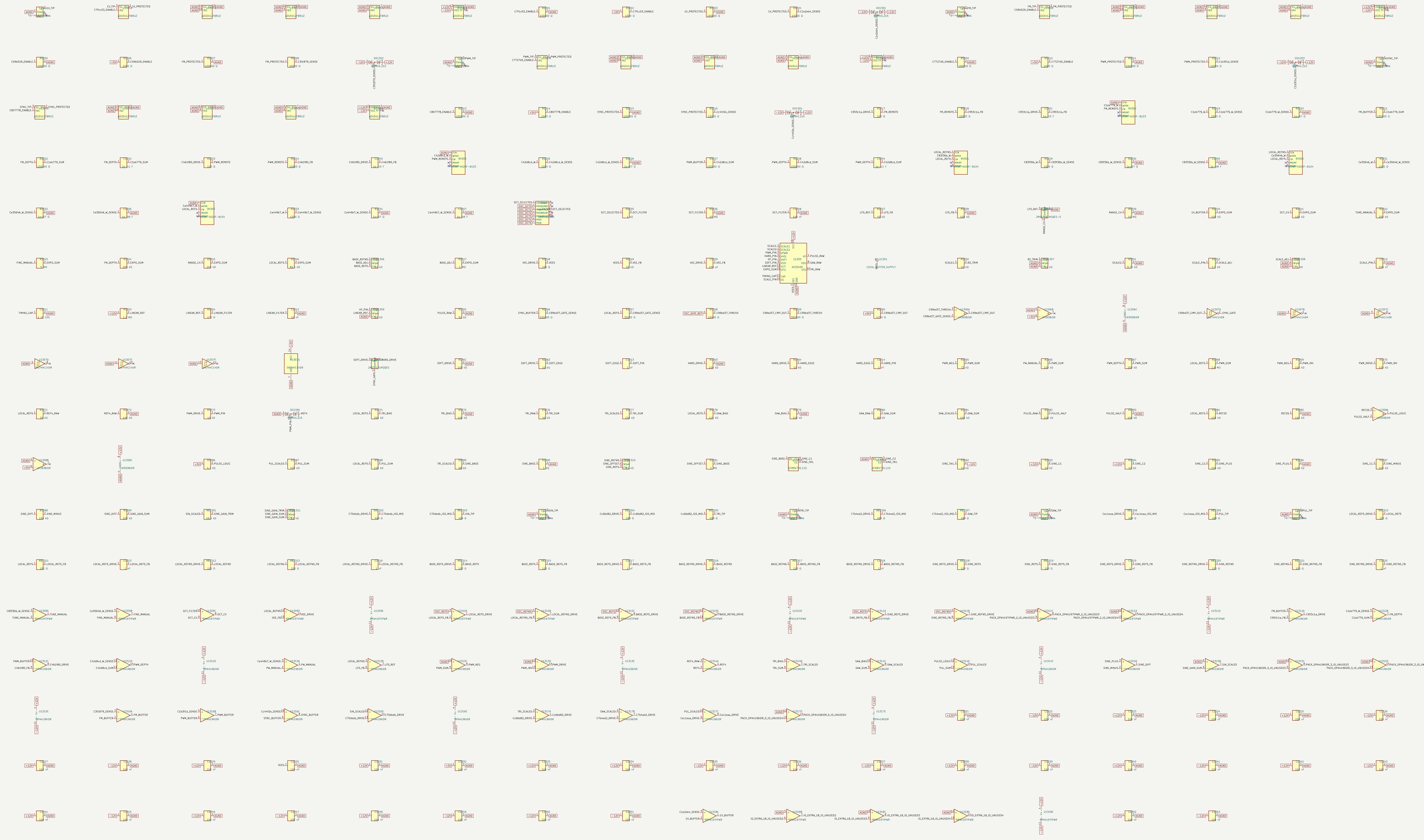
KiCad E.D.A. 10.0.6 | Id: 5/36



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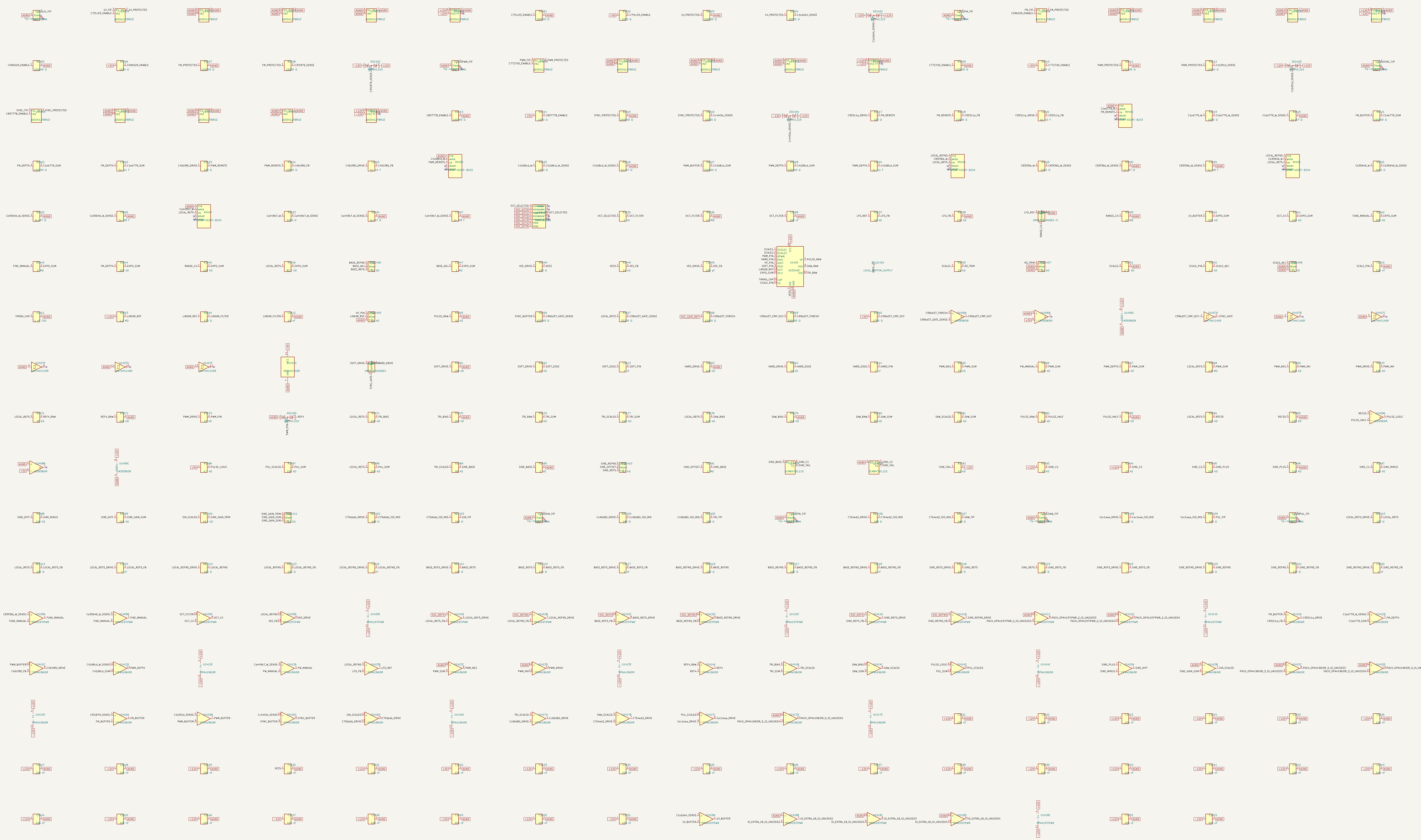
Size: A0 Date:
KiCad E.D.A. 10.0.6

Rev:
Id: 6/36



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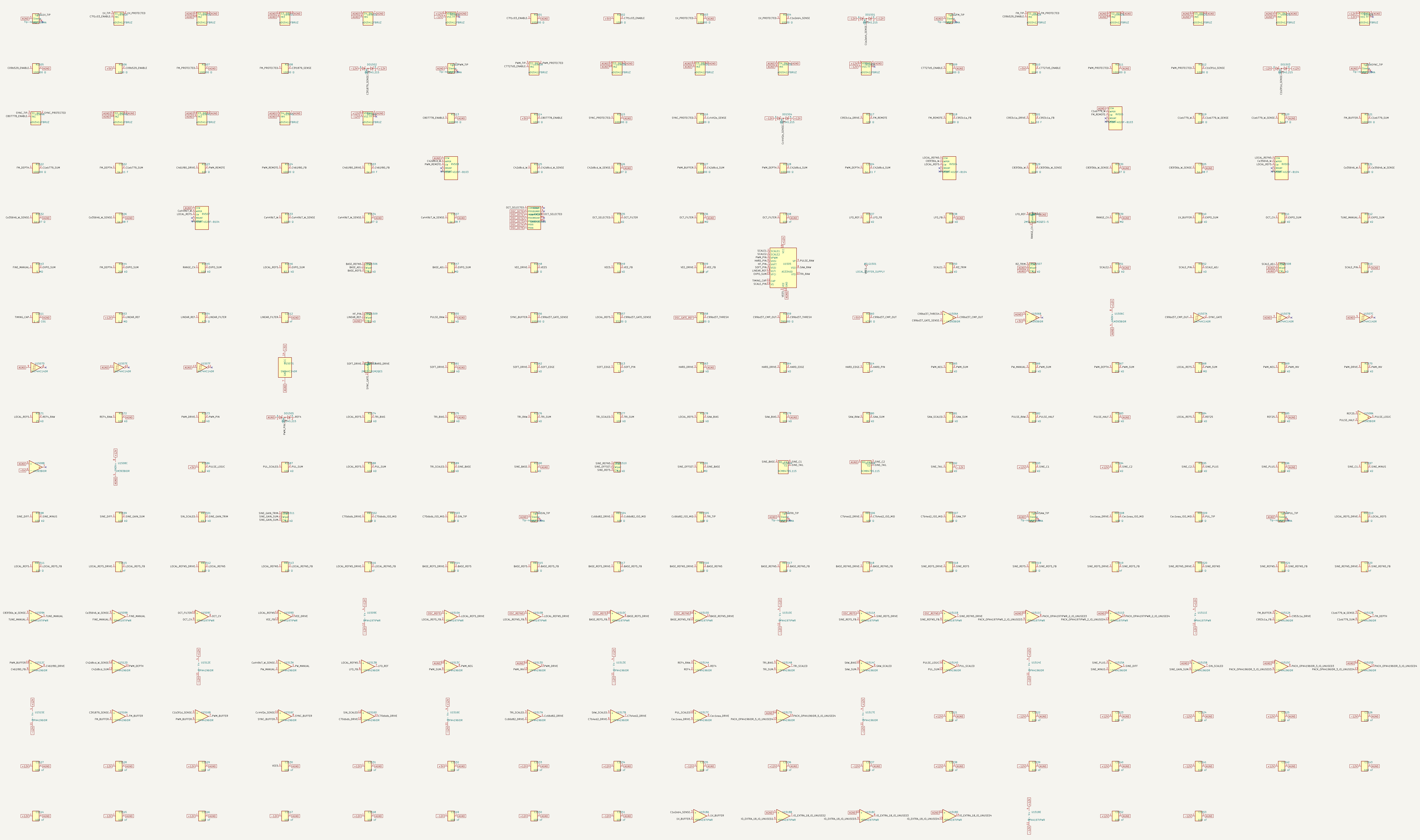
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/36



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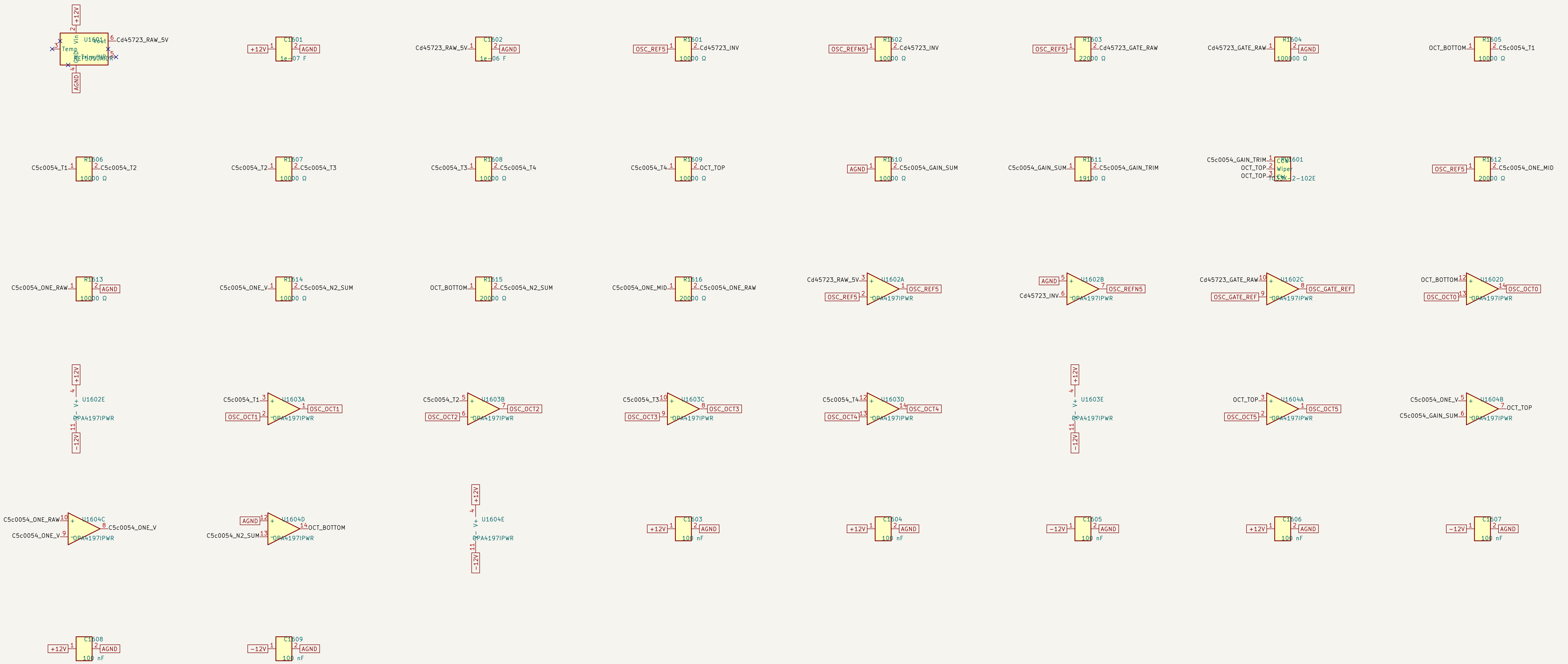
KiCad E.D.A. 10.0.6 | Id: 8/36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Vin
Vout
Temp
Trim/NR
GND
+12V
AGND
Cd45723_RAW_5V
C1601
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C1602
1e-06 F
R1601
OSC_REF5
Cd45723_INV
10000 Ω
R1602
OSC_REFN5
R1603
22000 Ω
Cd45723_GATE_RAW
R1604
100000 Ω
R1605
OCT_BOTTOM
C5c0054_T1
R1606
C5c0054_T2
R1607
C5c0054_T3
R1608
C5c0054_T4
R1609
OCT_TOP
R1610
C5c0054_GAIN_SUM
R1611
19100 Ω
C5c0054_GAIN_TRIM
RV1601
CCW
Wiper
CW
TC33X-2-102E
R1612
20000 Ω
C5c0054_ONE_MID
R1613
C5c0054_ONE_RAW
R1614
C5c0054_ONE_V
C5c0054_N2_SUM
R1615
R1616
U1602A
U1602B
U1602C
U1602D
U1602E
OPA4197IPWR
OSC_GATE_REF
OSC_OCT0
-12V
U1603A
U1603B
U1603C
U1603D
U1603E
OSC_OCT1
OSC_OCT2
OSC_OCT3
OSC_OCT4
U1604A
U1604B
U1604C
U1604D
U1604E
OSC_OCT5
C1603
C1604
C1605
C1606
C1607
C1608
C1609
100 nF

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Title:

Size: A2 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 9/36

Sheet: /F1/
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Title:

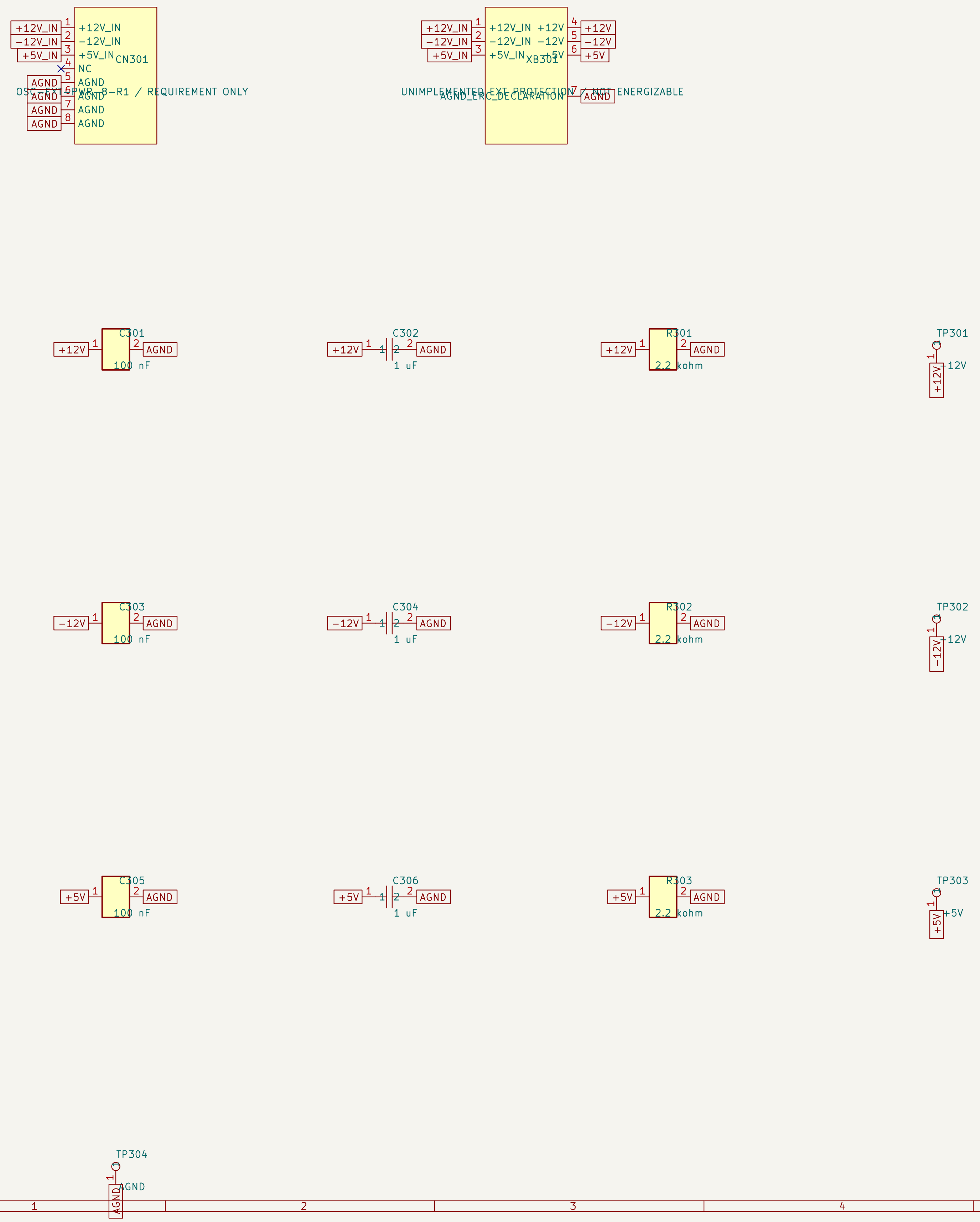
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/36

Sheet: /F2/
File: filter.kicad_sch
Title:
Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 11/36

Sheet: /F3/
File: filter.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 12/36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−12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
1
2
3
4
5
6
7
8
XB301
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
+12V_IN
−12V_IN
+5V_IN
+12V
−12V
+5V
AGND_ERC_DECLARATION
AGND
C301
100 nF
C302
1 uF
R301
2.2 kohm
TP301
+12V
C303
100 nF
C304
1 uF
R302
2.2 kohm
TP302
−12V
C305
100 nF
C306
1 uF
R303
2.2 kohm
TP303
+5V
TP304
AGND

Sheet: /POWER/
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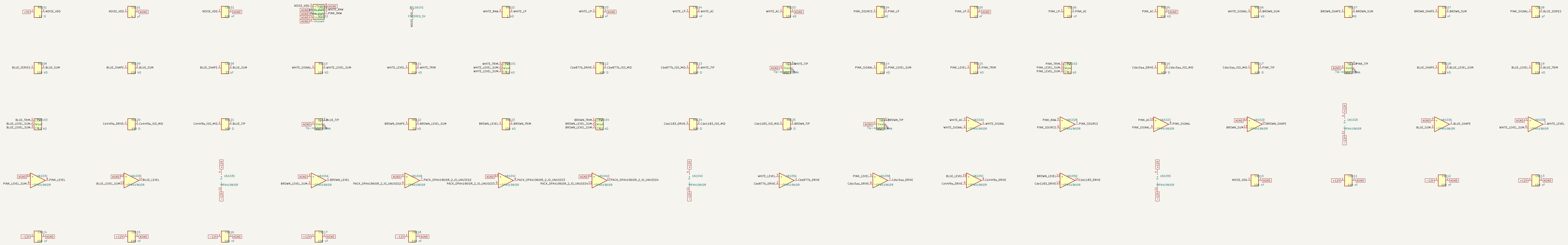
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/36


+5V R6101 NOISE_VDD 22 Ω
NOISE_VDD C6101 AGND 4.7 µF
NOISE_VDD C6102 AGND 100 nF
NOISE_VDD U6101 +5V0V AGND WHITE_RAW PINK_RAW NOISE2 Unused AGND
#FLG6101 NOISE_VDD FILTERED_5V
WHITE_RAW R6102 WHITE_LP 1 kΩ
WHITE_LP C6103 AGND 10 nF
WHITE_LP C6104 WHITE_AC 100 nF
WHITE_AC R6103 AGND 100 kΩ
PINK_SOURCE R6104 PINK_LP 1 kΩ
PINK_LP C6105 AGND 10 nF
PINK_LP C6106 PINK_AC 100 nF
PINK_AC R6105 AGND 100 kΩ
WHITE_SIGNAL R6106 BROWN_SUM 100 kΩ
BROWN_SHAPE R6107 BROWN_SUM 1 MΩ
BROWN_SHAPE C6107 BROWN_SUM 10 nF
PINK_SIGNAL C6108 BLUE_SERIES 100 pF
BLUE_SERIES R6108 BLUE_SUM 100 kΩ
BLUE_SHAPE R6109 BLUE_SUM 100 kΩ
BLUE_SHAPE C6109 BLUE_SUM 33 pF
WHITE_SIGNAL R6110 WHITE_LEVEL_SUM 100 kΩ
WHITE_LEVEL R6111 WHITE_TRIM 100 kΩ
WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM RV6101 CCW Wiper CW 10 kΩ
Cbe877b_DRIVE R6112 Cbe877b_ISO_MID 499 Ω
Cbe877b_ISO_MID R6113 WHITE_TIP 499 Ω
AGND S Tip Sleeve T WHITE_TIP TN Tip−normal
PINK_SIGNAL R6114 PINK_LEVEL_SUM 100 kΩ
PINK_LEVEL R6115 PINK_TRIM 100 kΩ
PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM RV6102 CCW Wiper CW 10 kΩ
Cdbc5aa_DRIVE R6116 Cdbc5aa_ISO_MID 499 Ω
Cdbc5aa_ISO_MID R6117 PINK_TIP 499 Ω
AGND S Tip Sleeve T PINK_TIP TN Tip−normal
BLUE_SHAPE R6118 BLUE_LEVEL_SUM 10 kΩ
BLUE_LEVEL R6119 BLUE_TRIM 100 kΩ
BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM RV6103 CCW Wiper CW 10 kΩ
Ce44f9a_DRIVE R6120 Ce44f9a_ISO_MID 499 Ω
Ce44f9a_ISO_MID R6121 BLUE_TIP 499 Ω
AGND S Tip Sleeve T BLUE_TIP TN Tip−normal
BROWN_SHAPE R6122 BROWN_LEVEL_SUM 20 kΩ
BROWN_LEVEL R6123 BROWN_TRIM 100 kΩ
BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM RV6104 CCW Wiper CW 10 kΩ
Cde1183_DRIVE R6124 Cde1183_ISO_MID 499 Ω
Cde1183_ISO_MID R6125 BROWN_TIP 499 Ω
AGND S Tip Sleeve T BROWN_TIP TN Tip−normal
WHITE_AC WHITE_SIGNAL U6102A WHITE_SIGNAL OPA4196IDR
PINK_RAW PINK_SOURCE U6102B PINK_SOURCE OPA4196IDR
PINK_AC PINK_SIGNAL U6102C PINK_SIGNAL OPA4196IDR
AGND BROWN_SUM U6102D BROWN_SHAPE OPA4196IDR
+12V U6102E V+ V− OPA4196IDR −12V
AGND BLUE_SUM U6103A BLUE_SHAPE OPA4196IDR
AGND WHITE_LEVEL_SUM U6103B WHITE_LEVEL OPA4196IDR
AGND PINK_LEVEL_SUM U6103C PINK_LEVEL OPA4196IDR
AGND BLUE_LEVEL_SUM U6103D BLUE_LEVEL OPA4196IDR
+12V U6103E V+ V− OPA4196IDR −12V
AGND BROWN_LEVEL_SUM U6104A BROWN_LEVEL OPA4196IDR
AGND PACK_OPA4196IDR_2_IO_UNUSED2 U6104B PACK_OPA4196IDR_2_IO_UNUSED2 OPA4196IDR
AGND PACK_OPA4196IDR_2_IO_UNUSED3 U6104C PACK_OPA4196IDR_2_IO_UNUSED3 OPA4196IDR
AGND PACK_OPA4196IDR_2_IO_UNUSED4 U6104D PACK_OPA4196IDR_2_IO_UNUSED4 OPA4196IDR
+12V U6104E V+ V− OPA4196IDR −12V
WHITE_LEVEL Cbe877b_DRIVE U6105A Cbe877b_DRIVE OPA4196IDR
PINK_LEVEL Cdbc5aa_DRIVE U6105B Cdbc5aa_DRIVE OPA4196IDR
BLUE_LEVEL Ce44f9a_DRIVE U6105C Ce44f9a_DRIVE OPA4196IDR
BROWN_LEVEL Cde1183_DRIVE U6105D Cde1183_DRIVE OPA4196IDR
+12V U6105E V+ V− OPA4196IDR −12V
NOISE_VDD C6110 AGND 100 nF
+12V C6111 AGND 100 nF
−12V C6112 AGND 100 nF
+12V C6113 AGND 100 nF
−12V C6114 AGND 100 nF
+12V C6115 AGND 100 nF
−12V C6116 AGND 100 nF
+12V C6117 AGND 100 nF
−12V C6118 AGND 100 nF

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Title:

Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 14/36

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Title:
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 15/36

Sheet: /E2/
File: envelope.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 16/36

Sheet: /E3/
File: envelope.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 17/36

Sheet: /E4/
File: envelope.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 18/36

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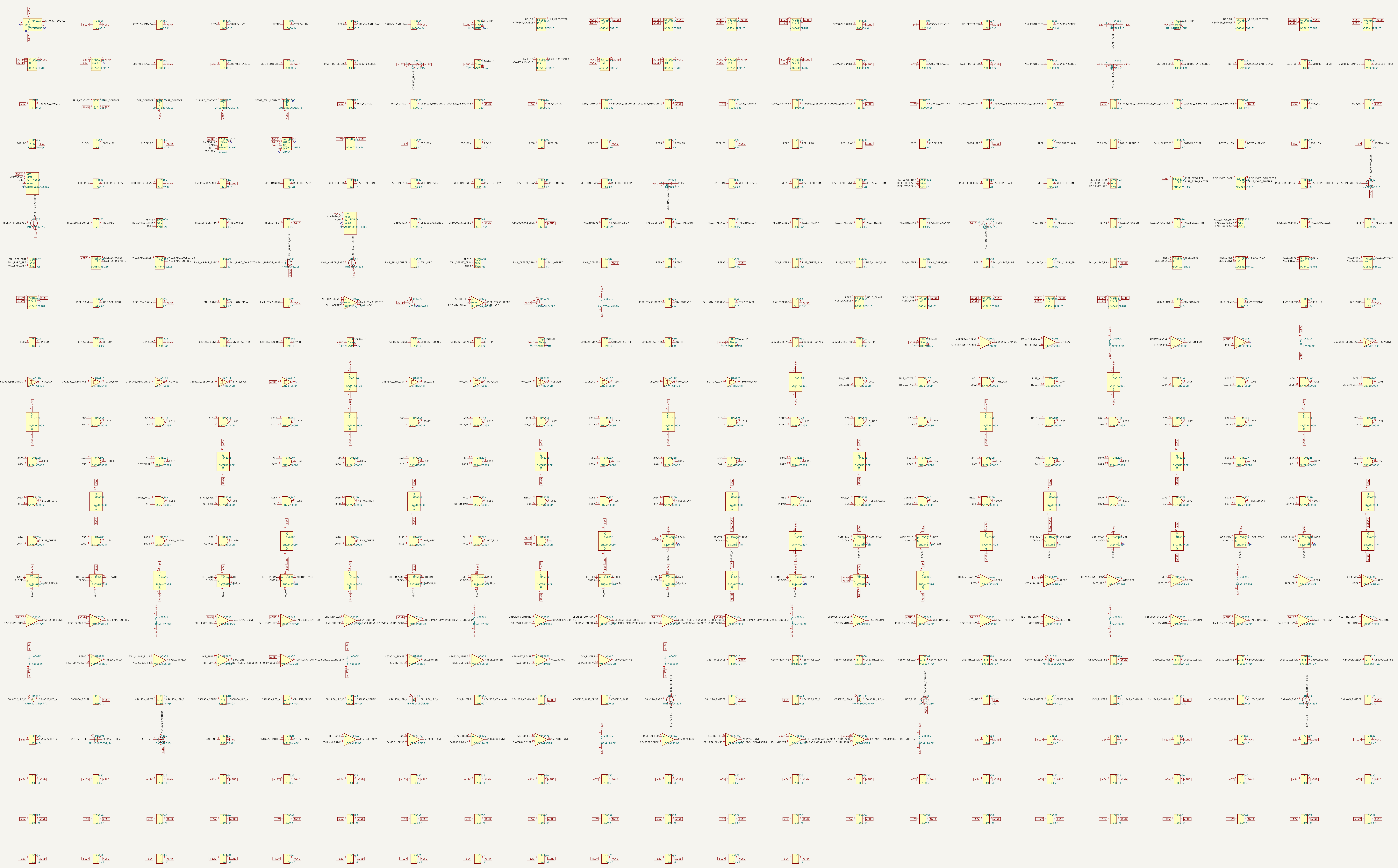
Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 19/36



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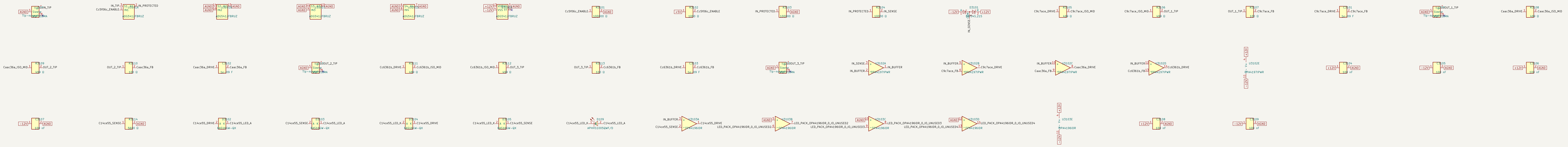
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 20/36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Tip
Sleeve
TN
Tip-normal
IN_TIP
Cc5f06c_ENABLE
U3101A
S1
D1
IN1
IN_PROTECTED
ADG5412FBRUZ
U3101B
S2
D2
IN2
ADG5412FBRUZ
U3101C
S3
D3
IN3
ADG5412FBRUZ
U3101D
S4
D4
IN4
ADG5412FBRUZ
+12V
-12V
VDD
GND
VSS
FF
ADG5412FBRUZ
R3101
100000 Ω
R3102
+5V
1000 Ω
R3103
100000 Ω
R3104
IN_SENSE
10000 Ω
D3101
BAT54S,215
R3105
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3106
OUT_1_TIP
499 Ω
R3107
C9c7ace_FB
100 Ω
C3101
1e-09 F
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3109
OUT_2_TIP
499 Ω
R3110
Caac36a_FB
100 Ω
C3102
1e-09 F
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3112
OUT_3_TIP
499 Ω
R3113
Cc6361b_FB
100 Ω
C3103
1e-09 F
U3102A
IN_BUFFER
OPA4197IPWR
U3102B
OPA4197IPWR
U3102C
OPA4197IPWR
U3102D
OPA4197IPWR
U3102E
V+
V-
OPA4197IPWR
C3104
100 nF
C3105
100 nF
C3106
100 nF
C3107
100 nF
R3114
C14ce55_SENSE
5600 Ω
D3102
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3103
BAS16GW-QX
D3104
C14ce55_LED_K
BAS16GW-QX
D3105
BAS16GW-QX
D109
APHHS1005QWF/D
U3103A
OPA4196IDR
U3103B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
OPA4196IDR
U3103C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
U3103D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3103E
OPA4196IDR
C3108
100 nF
C3109
100 nF

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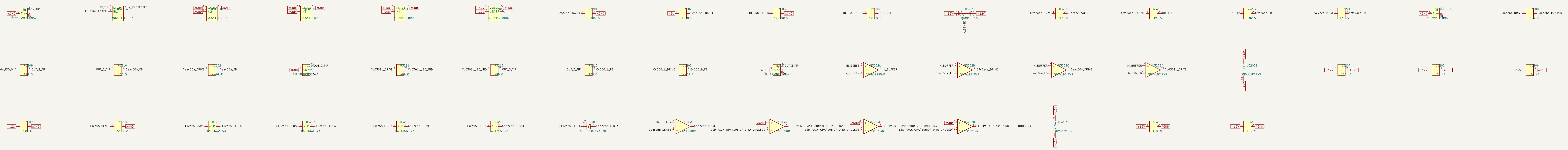
Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 21/36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Tip
Sleeve
Tip-normal
TN
IN_TIP
Cc5f06c_ENABLE
S1 D1
IN1
U3201A
IN_PROTECTED
ADG5412FBRUZ
S2 D2
IN2
U3201B
S3 D3
IN3
U3201C
S4 D4
IN4
U3201D
VDD GND
VSS FF
U3201E
+12V
-12V
R3201
Cc5f06c_ENABLE
100000 Ω
R3202
+5V
1000 Ω
R3203
IN_PROTECTED
100000 Ω
R3204
IN_SENSE
10000 Ω
D3201
BAT54S,215
IN_SENSE
R3205
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3206
OUT_1_TIP
499 Ω
R3207
C9c7ace_FB
100 Ω
C3201
1e-09 F
OUT_1_TIP
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
OUT_2_TIP
499 Ω
R3210
Caac36a_FB
100 Ω
C3202
1e-09 F
OUT_2_TIP
R3211
Cc6361b_DRIVE
Cc6361b_ISO_MID
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R3212
OUT_3_TIP
499 Ω
R3213
Cc6361b_FB
100 Ω
C3203
1e-09 F
OUT_3_TIP
U3202A
IN_SENSE
IN_BUFFER
OPA4197IPWR
U3202B
C9c7ace_FB
C9c7ace_DRIVE
U3202C
Caac36a_FB
Caac36a_DRIVE
U3202D
Cc6361b_FB
Cc6361b_DRIVE
U3202E
V+
V-
C3204
100 nF
C3205
100 nF
C3206
100 nF
C3207
100 nF
R3214
C14ce55_SENSE
5600 Ω
D3202
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3203
D3204
C14ce55_LED_K
D3205
D309
APHHS1005QWF/D
U3203A
OPA4196IDR
U3203B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3203C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3203D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3203E
C3208
100 nF
C3209
100 nF

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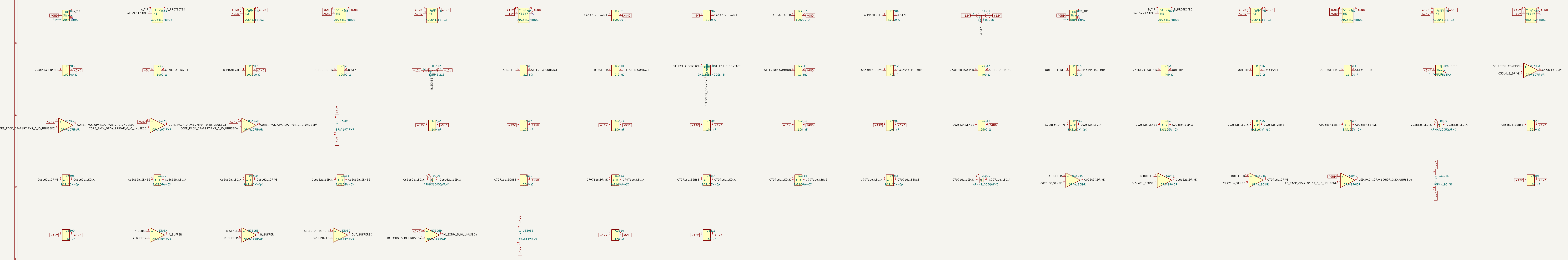
Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 22/36



AGND
A_TIP
Tip
Sleeve
Tip-normal
A_PROTECTED
Cadd797_ENABLE
S1
IN1
U3301A
ADG5412FBRUZ
S2
IN2
U3301B
S3
IN3
U3301C
S4
IN4
U3301D
+12V
-12V
VDD
VSS
U3301E
R3301
100000 Ω
+5V
R3302
1000 Ω
R3303
A_SENSE
R3304
D3301
BAV99
B_TIP
C9a8343_ENABLE
B_PROTECTED
R3305
R3306
R3307
R3308
B_SENSE
D3302
BAT54S,215
A_BUFFER
SELECT_A_CONTACT
R3309
2.2 kΩ
B_BUFFER
SELECT_B_CONTACT
R3310
SELECT_A_CONTACT
SELECT_B_CONTACT
2MS1T1B1M2QES-5
SELECTOR_COMMON
R3311
10 MΩ
C55d018_DRIVE
C55d018_ISO_MID
SELECTOR_REMOTE
OUT_BUFFERED
C61a19a_ISO_MID
OUT_TIP
C61a19a_FB
CORE_PACK_OPA4197IPWR_D_IO_UNUSED2
CORE_PACK_OPA4197IPWR_D_IO_UNUSED3
CORE_PACK_OPA4197IPWR_D_IO_UNUSED4
OPA4197IPWR
C025c3f_DRIVE
C025c3f_LED_A
C025c3f_SENSE
C025c3f_LED_K
APHHS1005QWF/D
Cc6c62b_SENSE
Cc6c62b_DRIVE
Cc6c62b_LED_A
Cc6c62b_LED_K
C7971de_SENSE
C7971de_DRIVE
C7971de_LED_A
C7971de_LED_K
OPA4196IDR
LED_PACK_OPA4196IDR_D_IO_UNUSED4
SELECTOR_REMOTE
C61a19a_FB
IO_EXTRA_S_IO_UNUSED4

Sheet: /X1/
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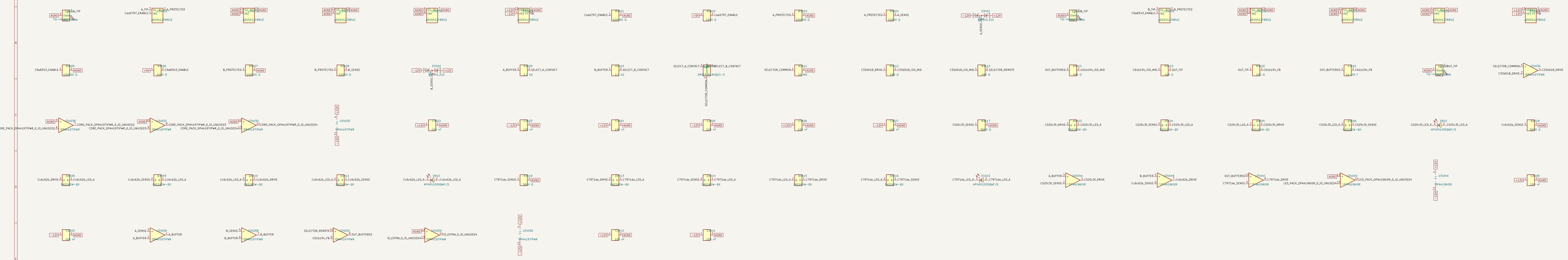
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 23/36



AGND
Tip
Sleeve
Tip-normal
A_TIP
A_TIP
Cadd797_ENABLE
U3401A
S1
D1
IN1
A_PROTECTED
ADG5412FBRUZ
U3401B
S2
D2
IN2
AGND
ADG5412FBRUZ
U3401C
S3
D3
IN3
AGND
ADG5412FBRUZ
U3401D
S4
D4
IN4
AGND
ADG5412FBRUZ
U3401E
+12V
-12V
VDD
VSS
GND
ADG5412FBRUZ
R3401
Cadd797_ENABLE
100000 Ω
AGND
R3402
+5V
1000 Ω
Cadd797_ENABLE
R3403
A_PROTECTED
100000 Ω
AGND
R3404
A_PROTECTED
10000 Ω
A_SENSE
D3401
-12V
+12V
BAT54S,215
A_SENSE
B_TIP
U3402A
B_TIP
C9a8343_ENABLE
B_PROTECTED
ADG5412FBRUZ
U3402B
U3402C
U3402D
U3402E
R3405
C9a8343_ENABLE
100000 Ω
AGND
R3406
+5V
1000 Ω
C9a8343_ENABLE
R3407
B_PROTECTED
100000 Ω
AGND
R3408
B_PROTECTED
10000 Ω
B_SENSE
D3402
-12V
+12V
BAT54S,215
B_SENSE
R3409
A_BUFFER
2.2 kΩ
SELECT_A_CONTACT
R3410
B_BUFFER
2.2 kΩ
SELECT_B_CONTACT
SELECT_A_CONTACT
SELECT_B_CONTACT
2MS1T1B1M2QES-5
SELECTOR_COMMON
R3411
SELECTOR_COMMON
10 MΩ
AGND
R3412
C33d018_DRIVE
499 Ω
C33d018_ISO_MID
R3413
C33d018_ISO_MID
499 Ω
SELECTOR_REMOTE
R3414
OUT_BUFFERED
499 Ω
C61b194_ISO_MID
R3415
C61b194_ISO_MID
499 Ω
OUT_TIP
R3416
OUT_TIP
100 Ω
C61b194_FB
C3401
OUT_BUFFERED
1e-09 F
C61b194_FB
OUT_TIP
U3403A
SELECTOR_COMMON
C33d018_DRIVE
C33d018_DRIVE
OPA4197IPWR
U3403B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
OPA4197IPWR
U3403C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
OPA4197IPWR
U3403D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
OPA4197IPWR
U3403E
OPA4197IPWR
C3402
+12V
100 nF
C3403
-12V
100 nF
C3404
+12V
100 nF
C3405
-12V
100 nF
C3406
+12V
100 nF
C3407
-12V
100 nF
R3417
C025c3f_SENSE
5600 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
C025c3f_SENSE
C025c3f_LED_A
BAS16GW-QX
D3405
C025c3f_LED_K
C025c3f_DRIVE
BAS16GW-QX
D3406
C025c3f_LED_K
C025c3f_SENSE
BAS16GW-QX
D810
C025c3f_LED_K
C025c3f_LED_A
APHHS1005QWF/D
R3418
Cc6c62b_SENSE
5600 Ω
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
BAS16GW-QX
D3409
Cc6c62b_SENSE
Cc6c62b_LED_A
BAS16GW-QX
D3410
Cc6c62b_LED_K
Cc6c62b_DRIVE
BAS16GW-QX
D3411
Cc6c62b_LED_K
Cc6c62b_SENSE
BAS16GW-QX
D910
Cc6c62b_LED_K
Cc6c62b_LED_A
APHHS1005QWF/D
R3419
C7971de_SENSE
5600 Ω
D3413
C7971de_DRIVE
C7971de_LED_A
BAS16GW-QX
D3414
C7971de_SENSE
C7971de_LED_A
BAS16GW-QX
D3415
C7971de_LED_K
C7971de_DRIVE
BAS16GW-QX
D3416
C7971de_LED_K
C7971de_SENSE
BAS16GW-QX
D1010
C7971de_LED_K
C7971de_LED_A
APHHS1005QWF/D
U3404A
A_BUFFER
C025c3f_SENSE
C025c3f_DRIVE
OPA4196IDR
U3404B
B_BUFFER
Cc6c62b_SENSE
Cc6c62b_DRIVE
OPA4196IDR
U3404C
OUT_BUFFERED
C7971de_SENSE
C7971de_DRIVE
OPA4196IDR
U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3404E
OPA4196IDR
C3408
+12V
100 nF
C3409
-12V
100 nF
U3405A
A_SENSE
A_BUFFER
OPA4197IPWR
U3405B
B_SENSE
B_BUFFER
OPA4197IPWR
U3405C
SELECTOR_REMOTE
C61b194_FB
OUT_BUFFERED
OPA4197IPWR
U3405D
IO_EXTRA_5_IO_UNUSED4
OPA4197IPWR
U3405E
OPA4197IPWR
C3410
+12V
100 nF
C3411
-12V
100 nF

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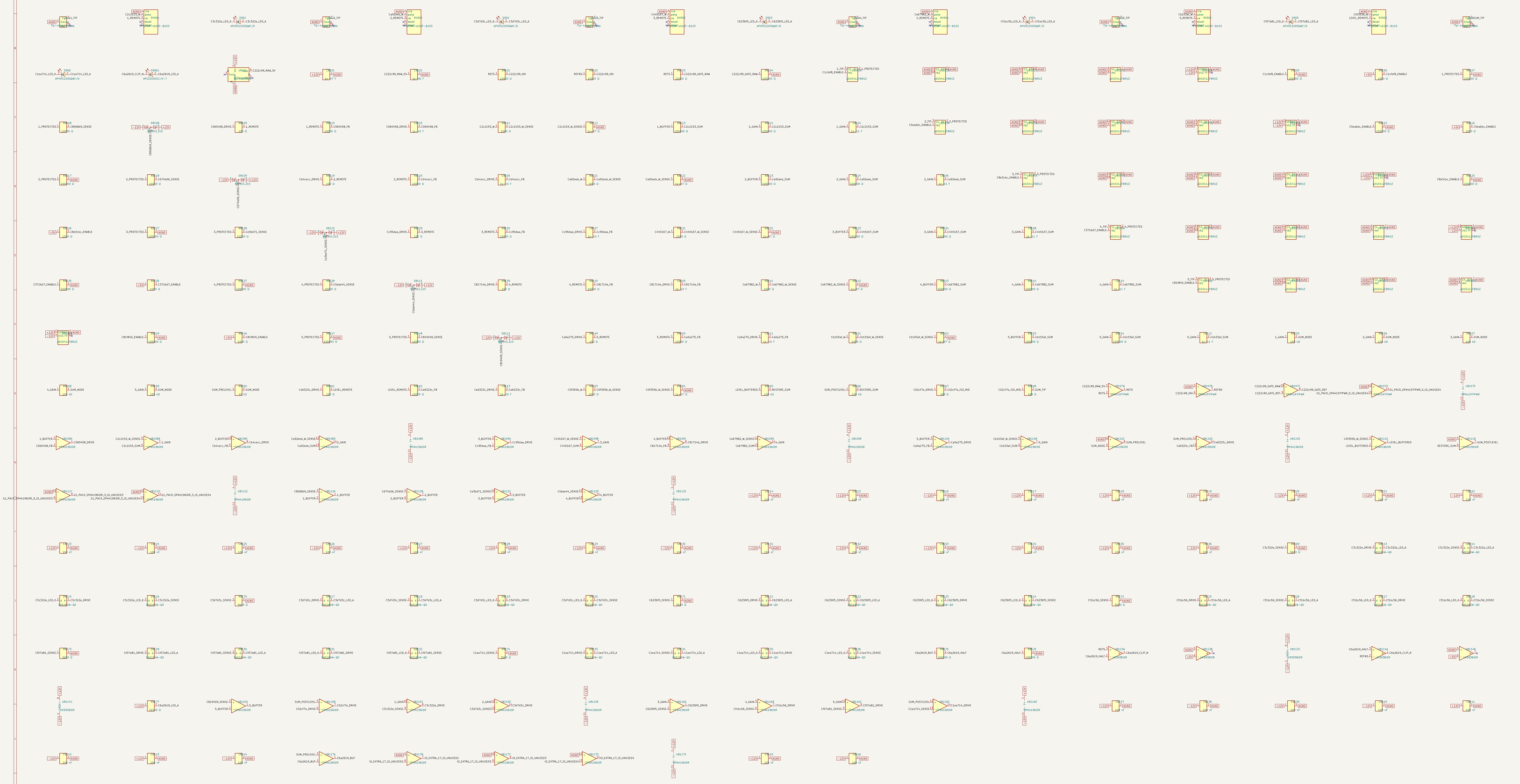
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 24/36



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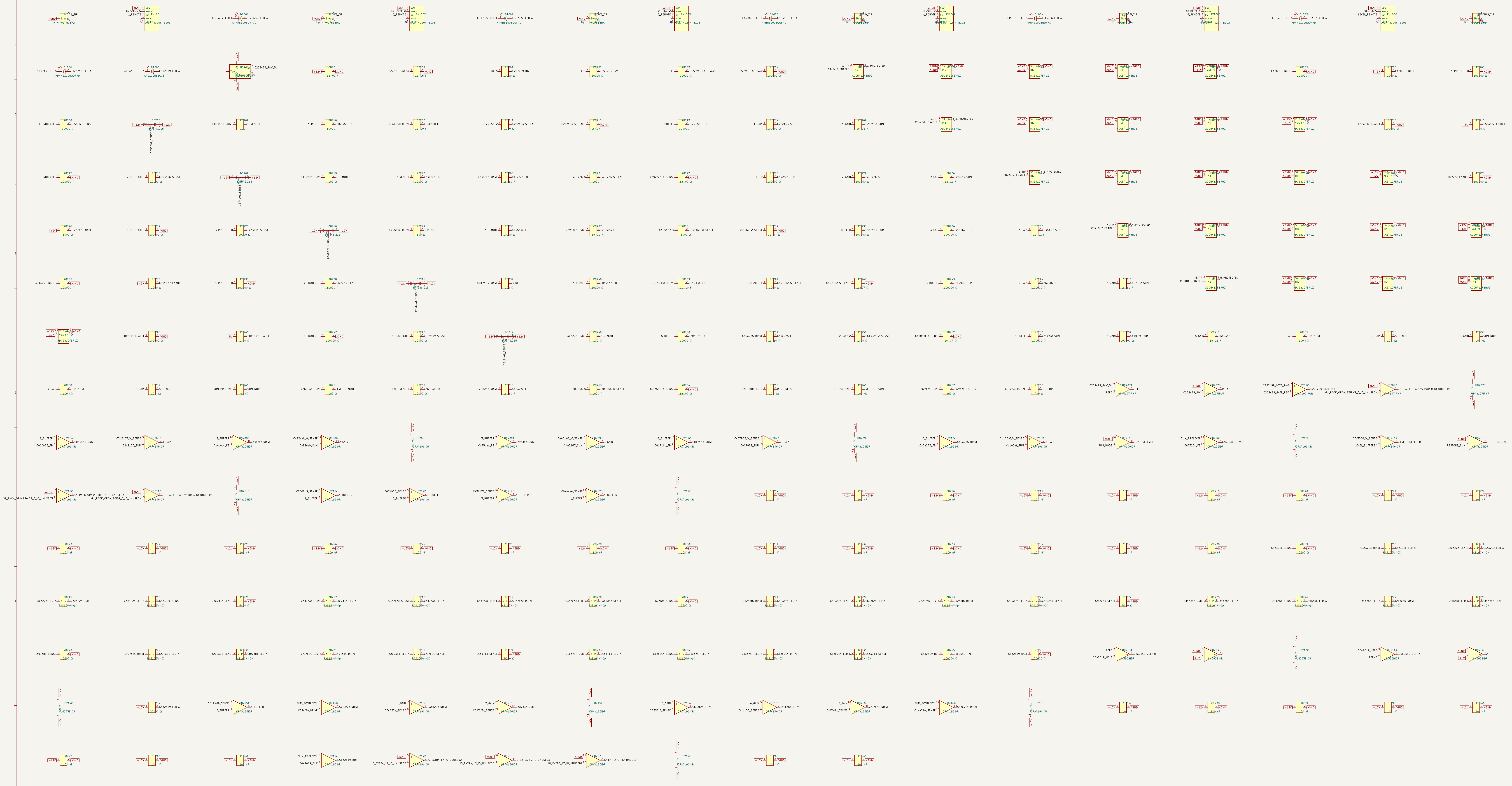
Id: 25/36



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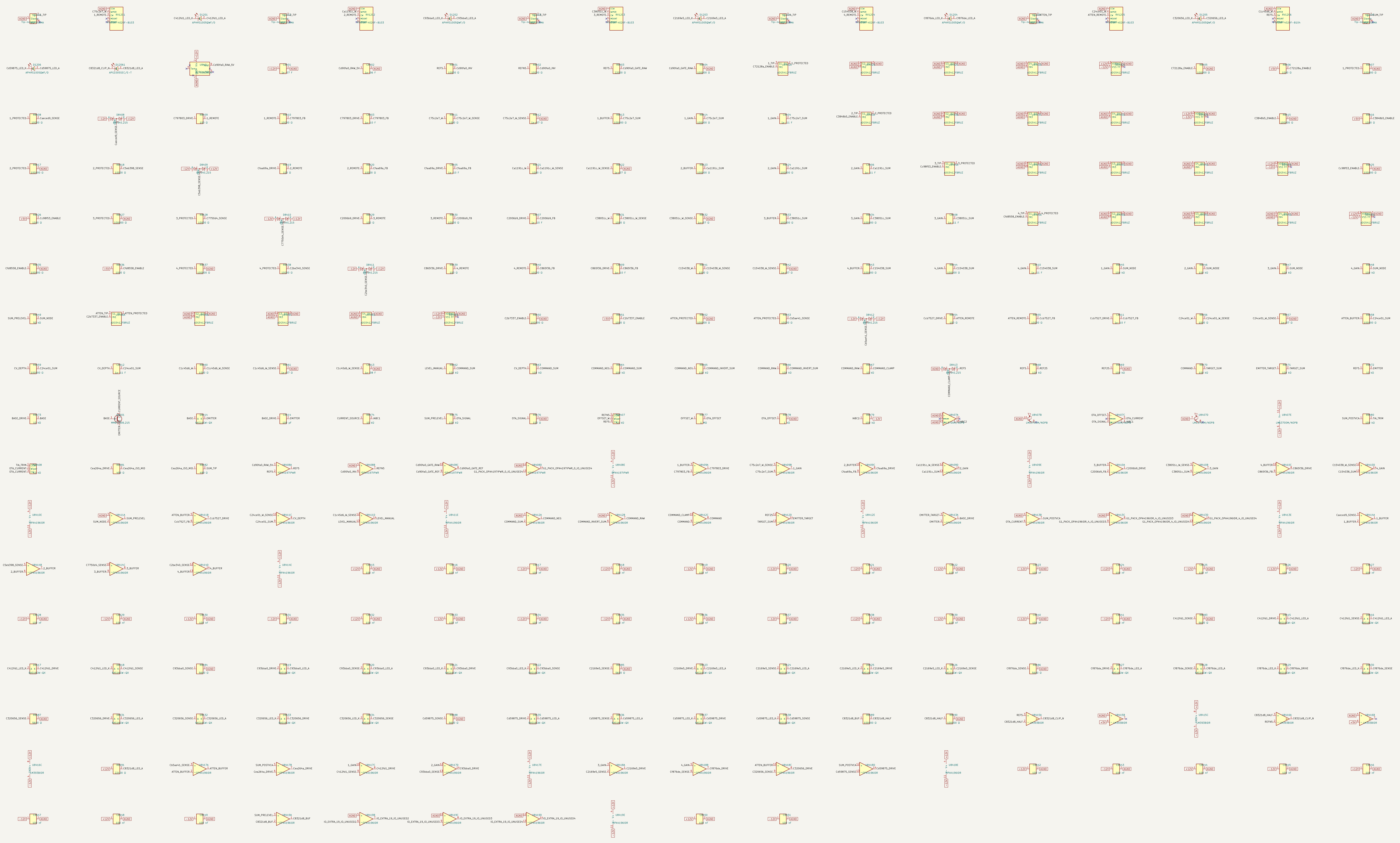
KiCad E.D.A. 10.0.6		Id: 27/36



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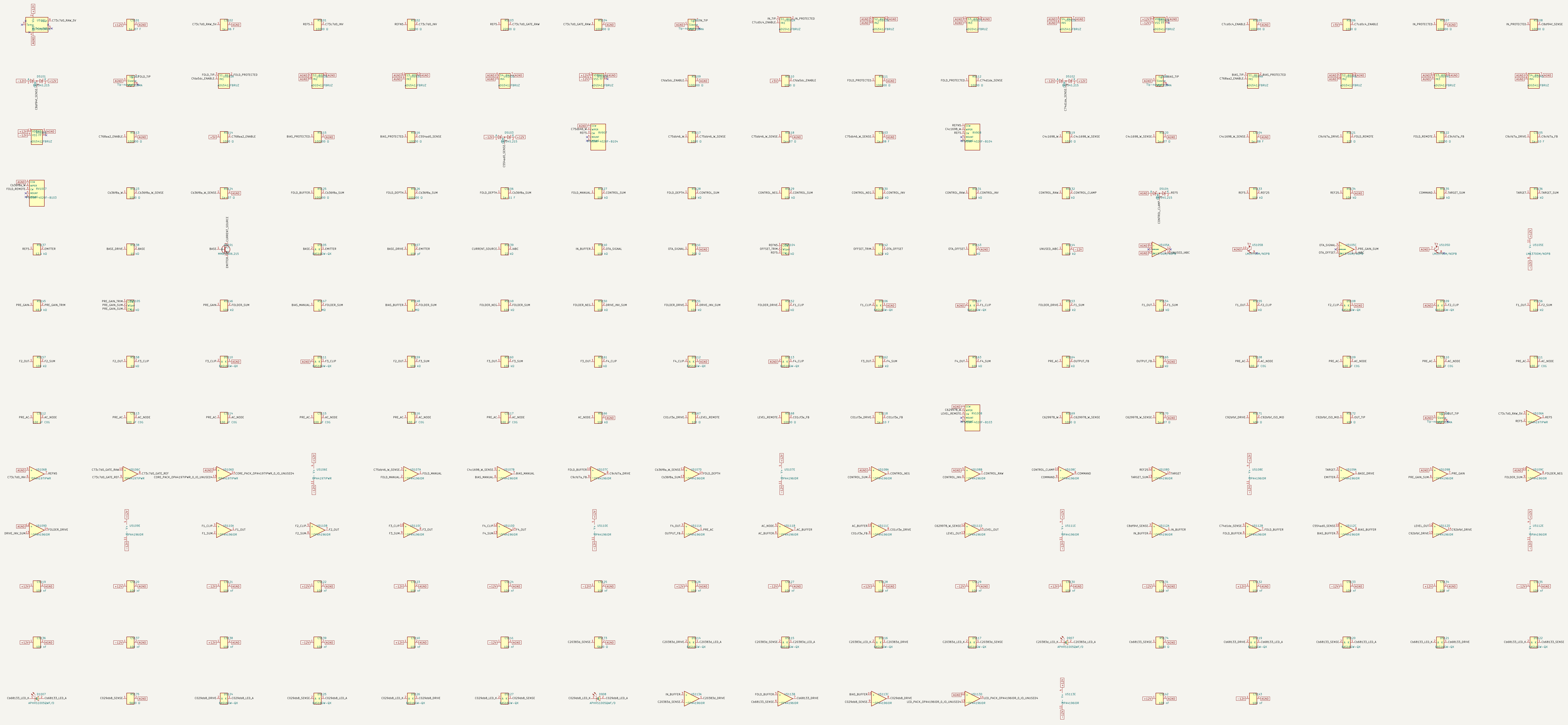
KiCad E.D.A. 10.0.6 | Id: 28/36



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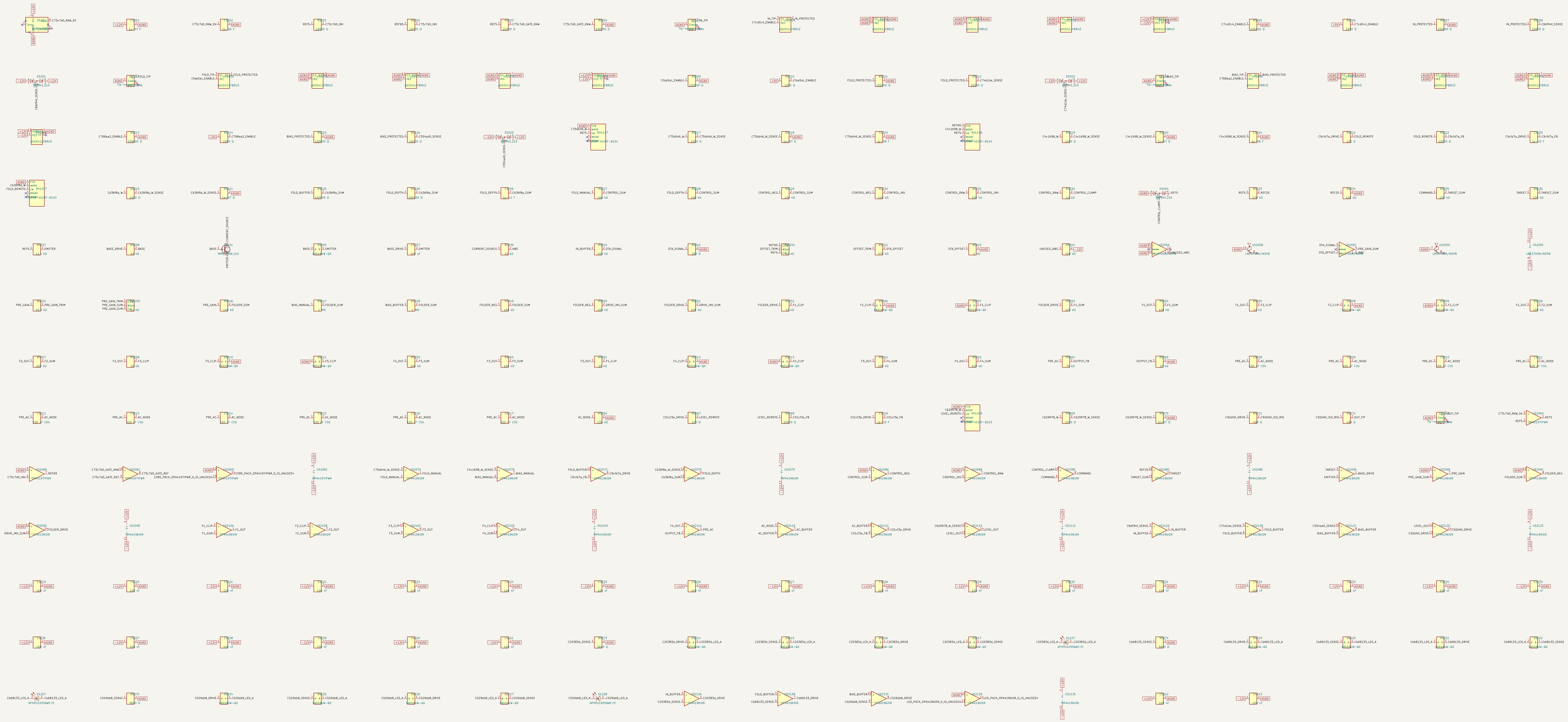

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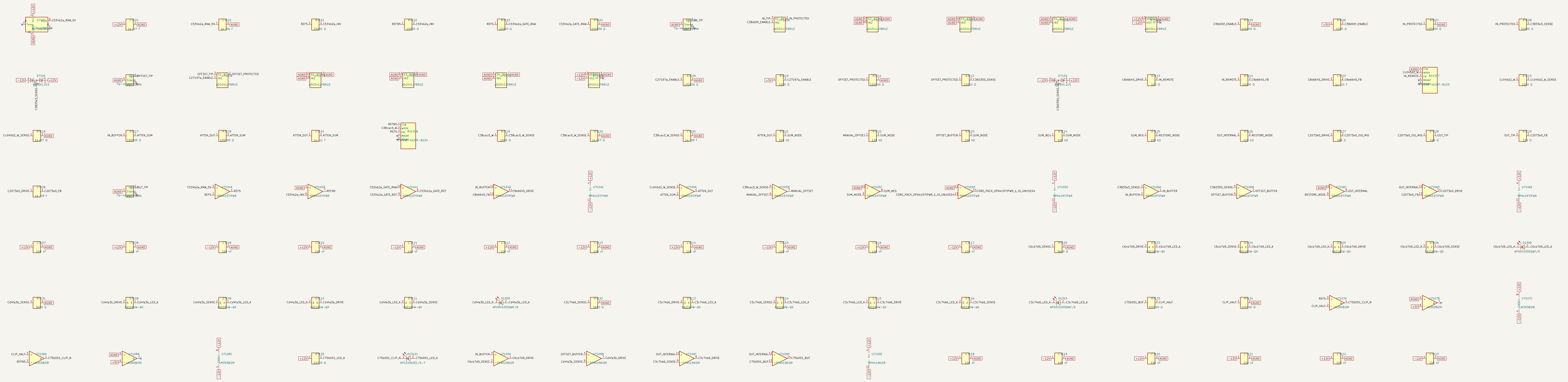

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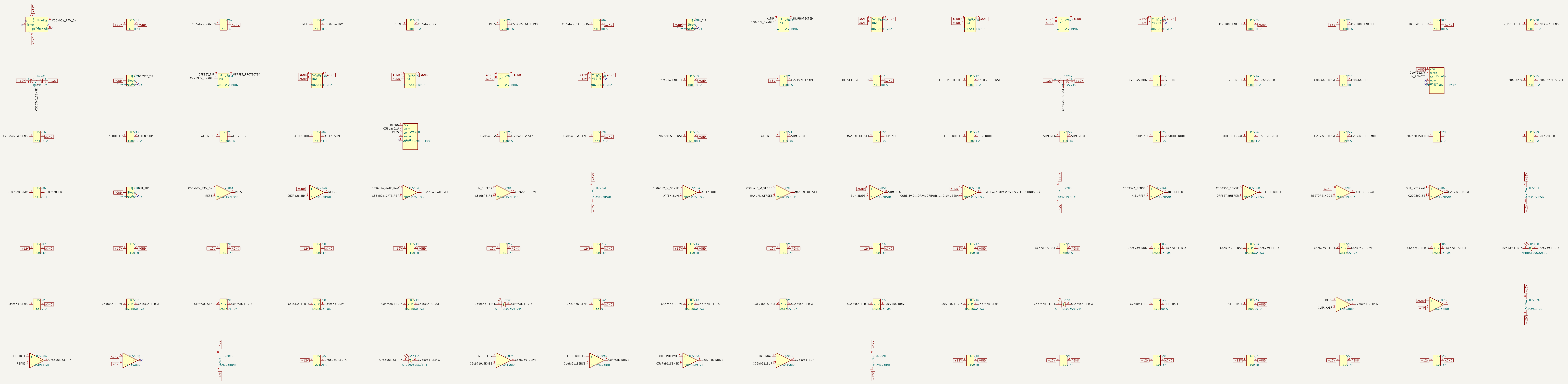
Id: 31/36



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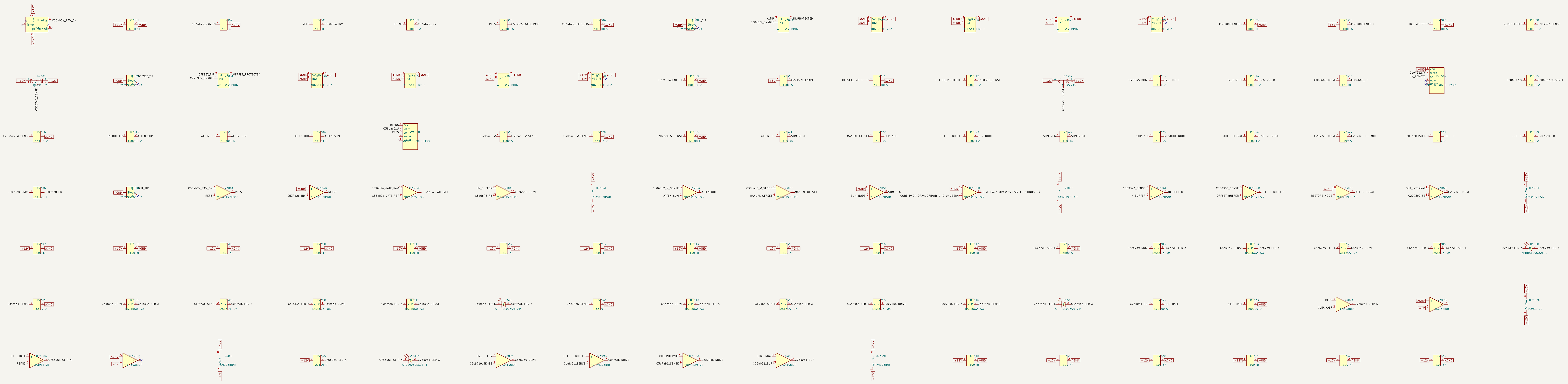

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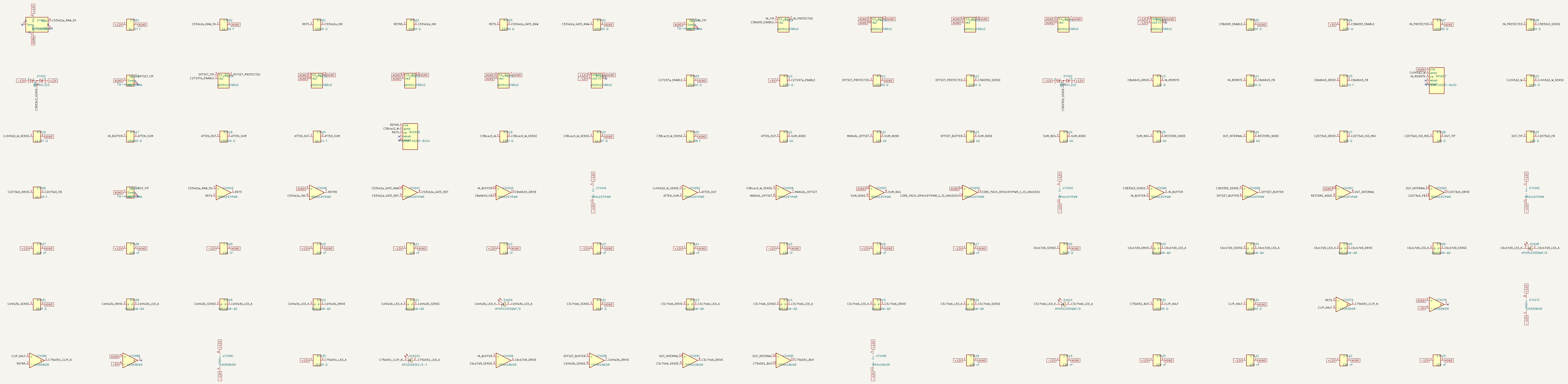

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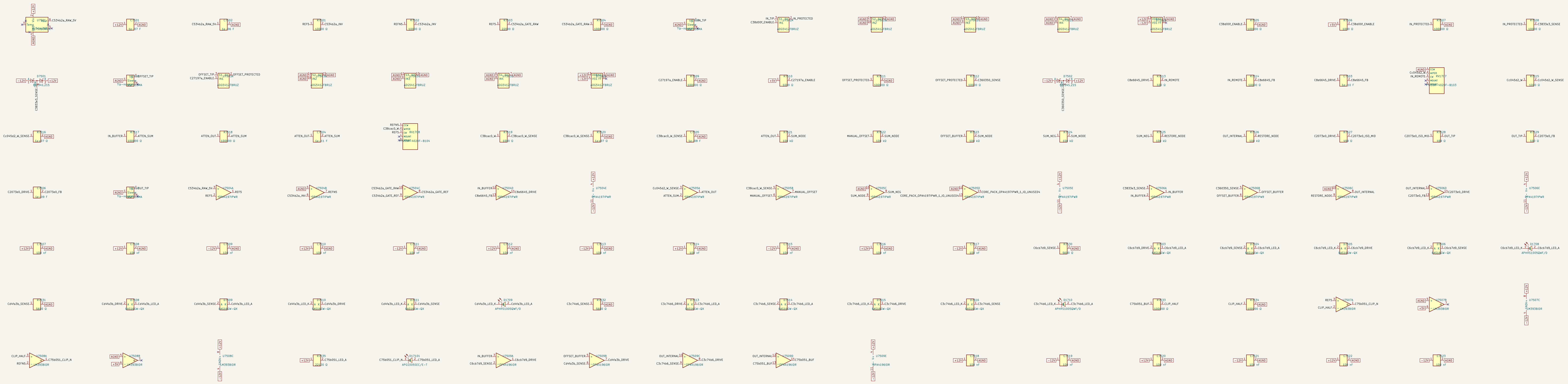
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C7501 +12V AGND 1e-07 F
C7502 C534b2a_RAW_5V AGND 1e-06 F
R7501 REF5 C534b2a_INV 10000 Ω
R7502 REFN5 C534b2a_INV 10000 Ω
R7503 REF5 C534b2a_GATE_RAW 22000 Ω
R7504 C534b2a_GATE_RAW AGND 100000 Ω
J1708 AGND Tip Sleeve IN_TIP
U7502A IN_TIP C38d00f_ENABLE IN_PROTECTED S1 D1 IN1 ADG5412FBRUZ
U7502B AGND AGND AGND S2 D2 IN2 ADG5412FBRUZ
U7502C AGND AGND AGND S3 D3 IN3 ADG5412FBRUZ
U7502D AGND AGND AGND S4 D4 IN4 ADG5412FBRUZ
U7502E +12V −12V VDD GND VSS AGND ADG5412FBRUZ
R7505 C38d00f_ENABLE AGND 100000 Ω
R7506 +5V C38d00f_ENABLE 1000 Ω
R7507 IN_PROTECTED AGND 100000 Ω
R7508 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7501 −12V +12V BAT54S,215 C3833e3_SENSE
J1709 AGND Tip Sleeve OFFSET_TIP
U7503A OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED S1 D1 IN1 ADG5412FBRUZ
U7503B AGND AGND AGND S2 D2 IN2 ADG5412FBRUZ
U7503C AGND AGND AGND S3 D3 IN3 ADG5412FBRUZ
U7503D AGND AGND AGND S4 D4 IN4 ADG5412FBRUZ
U7503E +12V −12V VDD GND VSS AGND ADG5412FBRUZ
R7509 C27197a_ENABLE AGND 100000 Ω
R7510 +5V C27197a_ENABLE 1000 Ω
R7511 OFFSET_PROTECTED AGND 100000 Ω
R7512 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7502 −12V +12V BAT54S,215 C360350_SENSE
R7513 C8e6645_DRIVE IN_REMOTE 100 Ω
R7514 IN_REMOTE C8e6645_FB 10000 Ω
C7503 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1707 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT PTV09A-4020F-B103
R7515 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7516 Cc045d2_W_SENSE AGND 1e+07 Ω
R7517 IN_BUFFER ATTEN_SUM 100000 Ω
R7518 ATTEN_OUT ATTEN_SUM 100000 Ω
C7504 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1708 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT PTV09A-4020F-B104
R7519 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7520 C38cac0_W_SENSE AGND 1e+07 Ω
C7505 C38cac0_W_SENSE AGND 1e-08 F
R7521 ATTEN_OUT SUM_NODE 100 kΩ
R7522 MANUAL_OFFSET SUM_NODE 100 kΩ
R7523 OFFSET_BUFFER SUM_NODE 100 kΩ
R7524 SUM_NEG SUM_NODE 100 kΩ
R7525 SUM_NEG RESTORE_NODE 100 kΩ
R7526 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7527 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7528 C2073e0_ISO_MID OUT_TIP 499 Ω
R7529 OUT_TIP C2073e0_FB 100 Ω
C7506 C2073e0_DRIVE C2073e0_FB 1e-09 F
J1710 AGND Tip Sleeve OUT_TIP
U7504A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7504B AGND C534b2a_INV REFN5 OPA4197IPWR
U7504C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7504D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7504E +12V −12V V+ V− OPA4197IPWR
U7505A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7505B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7505C AGND SUM_NODE SUM_NEG OPA4197IPWR
U7505D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IPWR
U7505E +12V −12V V+ V− OPA4197IPWR
U7506A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7506B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7506C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7506D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7506E +12V −12V V+ V− OPA4197IPWR
C7507 +12V AGND 100 nF
C7508 +12V AGND 100 nF
C7509 −12V AGND 100 nF
C7510 +12V AGND 100 nF
C7511 −12V AGND 100 nF
C7512 +12V AGND 100 nF
C7513 −12V AGND 100 nF
C7514 +12V AGND 100 nF
C7515 −12V AGND 100 nF
C7516 +12V AGND 100 nF
C7517 −12V AGND 100 nF
R7530 C6cb7d9_SENSE AGND 5600 Ω
D7503 C6cb7d9_DRIVE C6cb7d9_LED_A A K BAS16GW-QX
D7504 C6cb7d9_SENSE C6cb7d9_LED_A A K BAS16GW-QX
D7505 C6cb7d9_LED_K C6cb7d9_DRIVE A K BAS16GW-QX
D7506 C6cb7d9_LED_K C6cb7d9_SENSE A K BAS16GW-QX
D1708 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7531 Cd4fa3b_SENSE AGND 5600 Ω
D7508 Cd4fa3b_DRIVE Cd4fa3b_LED_A A K BAS16GW-QX
D7509 Cd4fa3b_SENSE Cd4fa3b_LED_A A K BAS16GW-QX
D7510 Cd4fa3b_LED_K Cd4fa3b_DRIVE A K BAS16GW-QX
D7511 Cd4fa3b_LED_K Cd4fa3b_SENSE A K BAS16GW-QX
D1709 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7532 C3c74b6_SENSE AGND 5600 Ω
D7513 C3c74b6_DRIVE C3c74b6_LED_A A K BAS16GW-QX
D7514 C3c74b6_SENSE C3c74b6_LED_A A K BAS16GW-QX
D7515 C3c74b6_LED_K C3c74b6_DRIVE A K BAS16GW-QX
D7516 C3c74b6_LED_K C3c74b6_SENSE A K BAS16GW-QX
D1710 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7533 C75b051_BUF CLIP_HALF 100000 Ω
R7534 CLIP_HALF AGND 100000 Ω
U7507A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7507B AGND +5V LM393BIDR
U7507C +12V −12V V+ GND LM393BIDR
U7508A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7508B AGND +5V LM393BIDR
U7508C +12V −12V V+ GND LM393BIDR
R7535 +12V C75b051_LED_A 22100 Ω
D17101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7509A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7509B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7509C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7509D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7509E +12V −12V V+ V− OPA4196IDR
C7518 +12V AGND 100 nF
C7519 −12V AGND 100 nF
C7520 +12V AGND 100 nF
C7521 −12V AGND 100 nF
C7522 +12V AGND 100 nF
C7523 −12V AGND 100 nF

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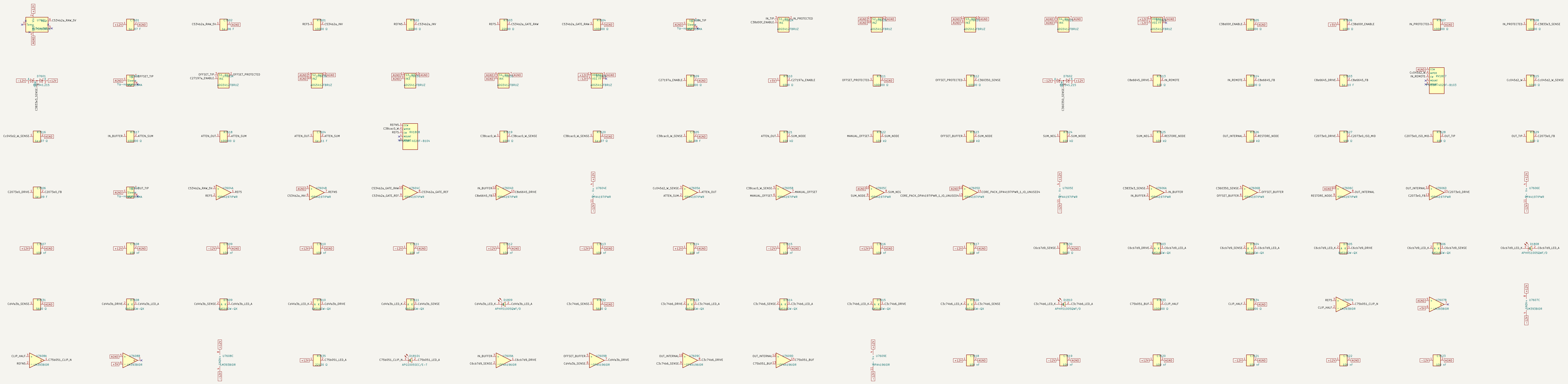
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 35/36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C7601 +12V 1e-07 F AGND
C7602 C534b2a_RAW_5V 1e-06 F AGND
R7601 REF5 10000 Ω C534b2a_INV
R7602 REFN5 10000 Ω C534b2a_INV
R7603 REF5 22000 Ω C534b2a_GATE_RAW
R7604 C534b2a_GATE_RAW 100000 Ω AGND
AGND S Tip Sleeve T TN IN_TIP
U7602A IN_TIP C38d00f_ENABLE S1 D1 IN1 IN_PROTECTED ADG5412FBRUZ
U7602B AGND AGND S2 D2 IN2 AGND ADG5412FBRUZ
U7602C AGND AGND S3 D3 IN3 AGND ADG5412FBRUZ
U7602D AGND AGND S4 D4 IN4 AGND ADG5412FBRUZ
U7602E +12V -12V VDD GND VSS FF AGND ADG5412FBRUZ
R7605 C38d00f_ENABLE 100000 Ω AGND
R7606 +5V 1000 Ω C38d00f_ENABLE
R7607 IN_PROTECTED 100000 Ω AGND
R7608 IN_PROTECTED 10000 Ω C3833e3_SENSE
D7601 -12V A K +12V BAT54S,215 COM C3833e3_SENSE
AGND S Tip Sleeve T TN OFFSET_TIP
U7603A OFFSET_TIP C27197a_ENABLE S1 D1 IN1 OFFSET_PROTECTED ADG5412FBRUZ
U7603B AGND AGND S2 D2 IN2 AGND ADG5412FBRUZ
U7603C AGND AGND S3 D3 IN3 AGND ADG5412FBRUZ
U7603D AGND AGND S4 D4 IN4 AGND ADG5412FBRUZ
U7603E +12V -12V VDD GND VSS FF AGND ADG5412FBRUZ
R7609 C27197a_ENABLE 100000 Ω AGND
R7610 +5V 1000 Ω C27197a_ENABLE
R7611 OFFSET_PROTECTED 100000 Ω AGND
R7612 OFFSET_PROTECTED 10000 Ω C360350_SENSE
D7602 -12V A K +12V BAT54S,215 COM C360350_SENSE
R7613 C8e6645_DRIVE 100 Ω IN_REMOTE
R7614 IN_REMOTE 10000 Ω C8e6645_FB
C7603 C8e6645_DRIVE 1e-10 F C8e6645_FB
RV1807 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B103
R7615 Cc045d2_W 1000 Ω Cc045d2_W_SENSE
R7616 Cc045d2_W_SENSE 1e+07 Ω AGND
R7617 IN_BUFFER 100000 Ω ATTEN_SUM
R7618 ATTEN_OUT 100000 Ω ATTEN_SUM
C7604 ATTEN_OUT 1e-11 F ATTEN_SUM
RV1808 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT MOUNT PTV09A-4020F-B104
R7619 C38cac0_W 1000 Ω C38cac0_W_SENSE
R7620 C38cac0_W_SENSE 1e+07 Ω AGND
C7605 C38cac0_W_SENSE 1e-08 F AGND
R7621 ATTEN_OUT 100 kΩ SUM_NODE
R7622 MANUAL_OFFSET 100 kΩ SUM_NODE
R7623 OFFSET_BUFFER 100 kΩ SUM_NODE
R7624 SUM_NEG 100 kΩ SUM_NODE
R7625 SUM_NEG 100 kΩ RESTORE_NODE
R7626 OUT_INTERNAL 100 kΩ RESTORE_NODE
R7627 C2073e0_DRIVE 499 Ω C2073e0_ISO_MID
R7628 C2073e0_ISO_MID 499 Ω OUT_TIP
R7629 OUT_TIP 100 Ω C2073e0_FB
C7606 C2073e0_DRIVE 1e-09 F C2073e0_FB
AGND S Tip Sleeve T TN OUT_TIP
U7604A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7604B AGND C534b2a_INV REFN5 OPA4197IPWR
U7604C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7604D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7604E +12V V+ V- -12V OPA4197IPWR
U7605A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7605B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7605C AGND SUM_NODE SUM_NEG OPA4197IPWR
U7605D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IPWR
U7605E +12V V+ V- -12V OPA4197IPWR
U7606A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7606B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7606C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7606D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7606E +12V V+ V- -12V OPA4197IPWR
C7607 +12V 100 nF AGND
C7608 +12V 100 nF AGND
C7609 -12V 100 nF AGND
C7610 +12V 100 nF AGND
C7611 -12V 100 nF AGND
C7612 +12V 100 nF AGND
C7613 -12V 100 nF AGND
C7614 +12V 100 nF AGND
C7615 -12V 100 nF AGND
C7616 +12V 100 nF AGND
C7617 -12V 100 nF AGND
R7630 C6cb7d9_SENSE 5600 Ω AGND
D7603 C6cb7d9_DRIVE A K C6cb7d9_LED_A BAS16GW-QX
D7604 C6cb7d9_SENSE A K C6cb7d9_LED_A BAS16GW-QX
D7605 C6cb7d9_LED_K A K C6cb7d9_DRIVE BAS16GW-QX
D7606 C6cb7d9_LED_K A K C6cb7d9_SENSE BAS16GW-QX
D1808 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7631 Cd4fa3b_SENSE 5600 Ω AGND
D7608 Cd4fa3b_DRIVE A K Cd4fa3b_LED_A BAS16GW-QX
D7609 Cd4fa3b_SENSE A K Cd4fa3b_LED_A BAS16GW-QX
D7610 Cd4fa3b_LED_K A K Cd4fa3b_DRIVE BAS16GW-QX
D7611 Cd4fa3b_LED_K A K Cd4fa3b_SENSE BAS16GW-QX
D1809 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7632 C3c74b6_SENSE 5600 Ω AGND
D7613 C3c74b6_DRIVE A K C3c74b6_LED_A BAS16GW-QX
D7614 C3c74b6_SENSE A K C3c74b6_LED_A BAS16GW-QX
D7615 C3c74b6_LED_K A K C3c74b6_DRIVE BAS16GW-QX
D7616 C3c74b6_LED_K A K C3c74b6_SENSE BAS16GW-QX
D1810 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7633 C75b051_BUF 100000 Ω CLIP_HALF
R7634 CLIP_HALF 100000 Ω AGND
U7607A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7607B AGND +5V LM393BIDR
U7607C +12V V+ GND -12V LM393BIDR
U7608A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7608B AGND +5V LM393BIDR
U7608C +12V V+ GND -12V LM393BIDR
R7635 +12V 22100 Ω C75b051_LED_A
D18101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7609A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7609B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7609C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7609D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7609E +12V V+ V- -12V OPA4196IDR
C7618 +12V 100 nF AGND
C7619 -12V 100 nF AGND
C7620 +12V 100 nF AGND
C7621 -12V 100 nF AGND
C7622 +12V 100 nF AGND
C7623 -12V 100 nF AGND

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